

End-to-End E-commerce Analytics Report

1. Project Overview

This project analyzes user behavior and business performance for an e-commerce platform using the **Google Analytics 4 sample dataset**. The goal is to extract actionable insights related to customer behavior, marketing channels, and revenue performance.

The analysis was performed using **Google BigQuery (SQL)** for data processing and **Looker Studio** for visualization.

Objectives

The main objectives of this project were to:

- Analyze core business metrics (revenue, orders, users)
- Measure ecommerce conversion performance
- Identify funnel drop-offs in the purchase journey
- Evaluate marketing channel effectiveness
- Analyze customer lifetime value
- Understand customer retention patterns using cohort analysis

2. Dataset

Source

“bigquery-public-data.ga4_obfuscated_sample_ecommerce.events_*”

This is a sample dataset of GA4 for Google Merchandise store for the period of November 2020 to Jan 2021 available in BigQuery, containing ecommerce events such as:

- session_start
- view_item
- add_to_cart
- begin_checkout
- purchase

Data Processing Steps

1. Raw GA4 event data was extracted.
2. Event parameters were flattened to extract key attributes.
3. Cleaned datasets were created for:
    - Sessions
    - Transactions
    - User summaries

These structured tables enabled efficient analytical queries.

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3. Data Architecture

The analysis pipeline followed this structure:

```
Raw GA4 Events
      ↓
Data Cleaning (BigQuery SQL)
      ↓
Session & Transaction Tables
      ↓
KPI / Funnel / Customer Analysis
```

```
       ↓
Business Insights & Dashboard
```

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4. Key Performance Indicators (KPI Analysis)

Core Metrics

Metric	Value
Total Revenue	$362,165
Total Orders	5,242
Total Users	268,023

The store generated **$362K in revenue from over 5,000 purchases**, while more than **268K users visited the platform**.

> This indicates strong user traffic but relatively **low purchasing activity**, suggesting opportunities to improve conversion.

Average Order Value (AOV)

Average Order Value measures how much revenue is generated per purchase.

Two AOV calculations were used because a **single session can contain multiple transactions**.

- **Session AOV:** $81.46
- **Transaction AOV:** $69.09

> The lower transaction-based AOV indicates that **some sessions include multiple purchases**, which increases the number of transactions relative to sessions.

Session Conversion Rate

💡 Only **1.4% of sessions resulted in a purchase**, meaning approximately **1 out of 70 sessions converts**. This highlights a significant opportunity to optimize the purchasing journey.

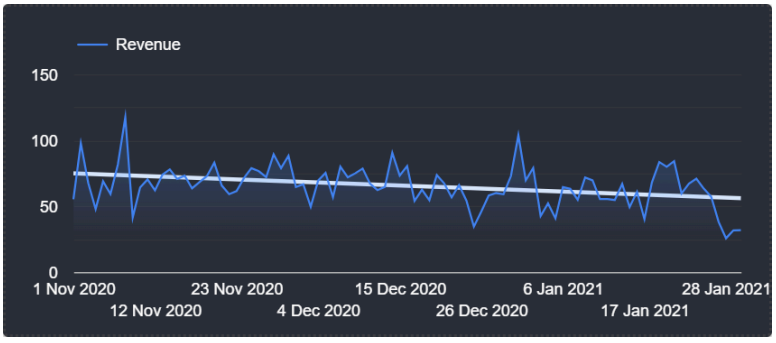
Repeat Purchase Rate

Approximately **18% of customers return to purchase again**, while the majority make only a single purchase.

Improving retention could significantly increase revenue.

Monthly Revenue Growth

Month	Revenue	Growth
Nov 2020	$144,260	—
Dec 2020	$160,555	+11%
Jan 2021	$57,350	−64%



> Revenue increased during the **holiday season in December**, followed by a **sharp drop in January**, suggesting seasonal purchasing behavior.

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5. Funnel Analysis

The ecommerce purchase journey consists of:

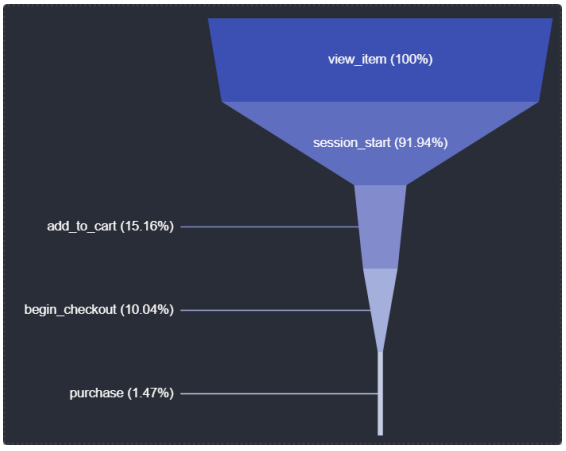
```
Product View → Add to Cart → Checkout → Purchase
```

Funnel Metrics

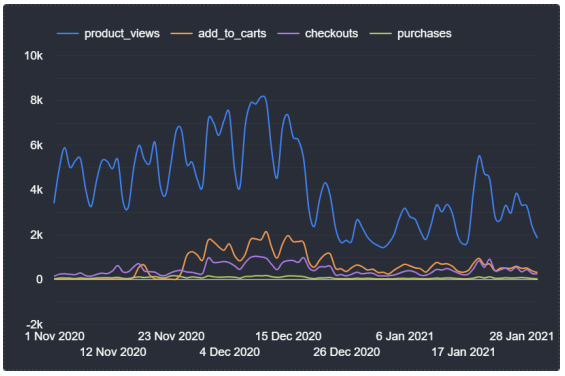
Stage	Sessions
Product Views	77,020
Add to Cart	15,188
Checkout	11,106
Purchase	4,848

Conversion Rates

Stage	Rate
View → Cart	19.7%
Cart → Checkout	73.1%
Checkout → Purchase	43.7%



Funnel Analysis



Stage trend over time

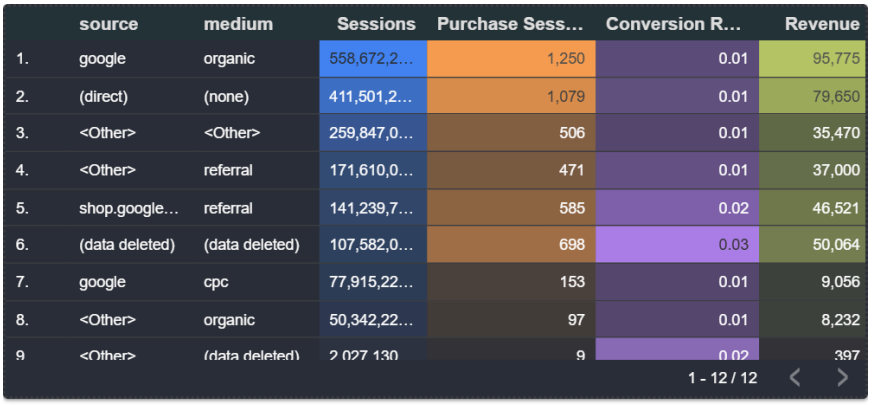
💡 **Large drop-off occurs between product views and cart additions.** Only **19.7% of users add products to the cart**, suggesting that product pages may not sufficiently encourage purchase intent. However, once items are added to the cart, most users proceed to checkout.

The **checkout stage shows a major drop-off**, with only **43% completing the purchase**, which may indicate friction such as:

- Shipping costs
- Checkout complexity
- Payment issues

6. Marketing Channel Performance

Traffic sources were analyzed to determine which marketing channels generate the most revenue.



	source	medium	Sessions	Purchase Sess...	Conversion R...	Revenue
1.	google	organic	558,672,2...	1,250	0.01	95,775
2.	(direct)	(none)	411,501,2...	1,079	0.01	79,650
3.	<Other>	<Other>	259,847,0...	506	0.01	35,470
4.	<Other>	referral	171,610,0...	471	0.01	37,000
5.	shop.google...	referral	141,239,7...	585	0.02	46,521
6.	(data deleted)	(data deleted)	107,582,0...	698	0.03	50,064
7.	google	cpc	77,915,22...	153	0.01	9,056
8.	<Other>	organic	50,342,22...	97	0.01	8,232
9	<Other>	(data deleted)	2,027,130	9	0.02	397

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Organic(unpaid) search via Google is the most important acquisition channel, generating the highest traffic and revenue.

Direct traffic(means ***the traffic source is unknown)*** also performs strongly, suggesting strong **brand recognition and returning users**.

Referral sources (means the traffic is originating from another website, excluding search engines (organic) or direct traffic) generate higher **revenue per session**, indicating high-intent users.

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7. Customer Lifetime Value (CLV)

Customer Lifetime Value measures the **total revenue generated by a customer during their relationship with the store**.

> The average customer contributes approximately **$89 over their lifetime**, slightly higher than the average order value. This suggests that most customers **make only one purchase**, while a smaller portion of users generate additional revenue through repeat purchases.

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8. CLV by Marketing Channel

Source	Medium	Avg CLV
<Other>	Organic	$157.36
<Data Deleted>	<Data Deleted>	$132.86
Google Merch Store	Referral	$127.65
Direct	None	$123.28
Google	Organic	$114.47
Google	CPC	$77.23

Organic and referral channels produce **higher-value customers**, while paid search users generate lower lifetime value.

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9. Cohort Retention Analysis

Customer cohorts were analyzed based on their **first purchase month**.

Month Since First Purchase	Active Users	Retention
November 2020	1,482	-
December 2020	391	26%
January 2021	105	7%

> Retention drops significantly after the first month. Only **~26% return in month one**, and **~7% remain active by month two**. This indicates opportunities to improve **customer retention strategies**.

10. RFM Analysis

RFM analysis segments customers based on **Recency (R), Frequency (F), and Monetary value (M)** to understand purchasing behavior and identify high-value users.



RFM segmentation

💡 Focusing on **retaining Champions, converting New Customers, and re-engaging At Risk customers** can significantly improve customer lifetime value and long-term revenue.

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11. Business Recommendations

Based on the analysis, the following actions could improve ecommerce performance:

Improve Product Page Engagement

Enhance product descriptions, reviews, and images to increase add-to-cart rates.

Reduce Checkout Friction

Simplify checkout steps and improve payment options to reduce abandonment.

Invest in High-Performing Channels

Organic search and referral traffic generate high-value customers.

Improve Customer Retention

Introduce loyalty programs, targeted promotions, and email campaigns to encourage repeat purchases.

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12. Tools Used

- **Google BigQuery** — SQL data processing
- **SQL** — Data transformation and analysis
- **Looker Studio** — Dashboard visualization
- **GitHub** — Project documentation and version control